KGL-IDS: Knowledge Graph-Based Lightweight Intrusion Detection for Industrial IoT Using Hybrid GWO-BPSO Feature Selection

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ABSTRACT

Intrusion detection systems (IDS) are key components of security in both conventional and Industrial Internet of Things (IIoT) networks. Recent machine learning- and deep learning-based IDS implementations have shifted from packet inspection to flow-based approaches. However, these methods suffer from three major limitations: they do not fully exploit relationships among traffic entities, they often overlook class imbalance, and the large number of features increases model complexity, limiting deployment in resource-constrained IIoT environments. To address these challenges, this work proposes KGL-IDS, a lightweight knowledge graph-driven IDS for IIoT edge environments that integrates semantic feature enhancement, data balancing, and hybrid feature reduction. Knowledge graph-based feature engineering captures relational patterns in traffic data, enriching the feature space. The Synthetic Minority Oversampling Technique (SMOTE) is then applied to balance class distribution. A hybrid feature selection method that combines Grey Wolf Optimization (GWO) and Binary Particle Swarm Optimization (BPSO) is employed to reduce dimensionality, thereby enabling efficient learning. Finally, a lightweight Artificial Neural Network (ANN) is used for attack detection and classification. The proposed approach is evaluated through ablation and comparative analysis on the WUSTL-IIoT and Edge-IIoT datasets. The results show that KGL-IDS achieves 99.62% and 99.19% accuracy using only 23 and 27 features, respectively. These findings demonstrate that the proposed framework provides an effective trade-off among accuracy, robustness, and computational efficiency for IIoT edge intrusion detection.

INTRODUCTION

The Internet of Things (IoT) has transformed technological applications across various sectors, including industrial domains, and continues to grow rapidly. The number of IoT devices has surpassed 20 billion in 2025 and is expected to reach around 32 billion by the end of 2030 Mallidi and Ramisetty (2025b), out of which approximately 45% belong to enterprise IoT. Although a precise standalone global count for Industrial IoT (IIoT) devices is less consistently reported, these figures indicate rapid growth in IIoT deployments as well. This expansion significantly increases the overall scale and complexity of interconnected systems, rendering traditional network security approaches insufficient for IoT/IIoT environments and necessitating their refinement.

This exponential growth has also led to a substantial rise in cyber threats, making IoT/IIoT devices and networks prime targets for attacks Choudhary (2024). Various reports indicate a more than 200% year-on-year increase in attacks on IoT devices over the past five years Jullian et al. (2023). One of the key reasons for this trend is the inherent challenges of IoT ecosystems, such as heterogeneity, dynamic topology, and large-scale connectivity, which limit the effectiveness of traditional security mechanisms

Rana et al. (2022). This creates a need for advanced and adaptive security solutions, including both cryptographic and detection-based approaches Arabi et al. (2026). Due to the high volume and velocity of traffic in IoT/IIoT networks, preventive cryptographic techniques must be lightweight while ensuring strong security. Techniques such as GIFT, designed for low-resource environments, demonstrate promising performance Yasmin and Gupta (2023), while EGCrypto leverages elliptic Galois encryption to improve performance under noisy conditions Kaur et al. (2023). In general, elliptic curve cryptography variants have dominated lightweight encryption approaches in IoT Adeniyi et al. (2024). However, in the emerging post-quantum era, the long-term reliability of many cryptographic techniques is being questioned, making them a complementary rather than a standalone solution for securing IoT/IIoT systems Schöffel et al. (2022).

On the other hand, detection approaches such as Intrusion Detection Systems (IDS) remain effective in identifying anomalies and intruders. The IDS sit at the perimeter or within a device and monitor incoming and outgoing traffic, stopping malicious traffic. These systems were broadly classified into signature-based and anomaly-based. Due to the performance of the anomaly-based IDS, there has been a significant shift towards them in recent years. These IDSs typically observe traffic behavior rather than merely inspecting packet headers Al-Ghuwairi et al. (2023). The integration of machine learning (ML) and deep learning (DL) into anomaly-IDS systems further enhanced their intrusion detection capabilities Xin et al. (2018). But these traditional IDSs are more resource-intensive, making them deployable only on high-resource cloud nodes in IoT/IIoT networks, not at the fog or edge layers. But this increases latency as the network grows; hence, it was recommended to move from cloud-centric to edge-centric approaches Kong et al. (2023); Xue et al. (2025). An IDS must be lightweight to be deployable on resource-constrained edge nodes, which has become very difficult to achieve Pérez et al. (2022).

Most existing anomaly-based IDS systems were built using flow-based features extracted from captured traffic datasets with tools such as Zeek, CICFlowMeter, and Argus. The datasets generated by these tools typically contain several features, ranging from 40 to 80, or more Mondragon et al. (2025). The CICFlowMeter efficiently generates 80 flow features, which are very promising for network traffic classification Ferriyan et al. (2021). Argus is much richer, providing over 175 flow features and numerous connection logs, whereas Zeek generates transaction logs that must be preprocessed to extract features Bagui et al. (2023). The drawbacks of these features are, first, that they capture statistical information at the flow level and typically ignore intricate underlying relationships among flows, and second, that some features are highly correlated and others are insignificant for the specific traffic classification task, such as anomaly detection Li et al. (2024).

In recent years, knowledge graph (KG) based feature extraction techniques have gained popularity for their ability to capture these underlying relationships Shang et al. (2024). But such graph-derived features alone are not sufficient for traffic classification, since effective intrusion detection also depends on preserving the discriminative information contained in conventional flow-level attributes. But using knowledge-graph-based features alongside the existing flow features further increases dimensionality, potentially introducing redundancy. Constructing an ML/DL-based IDS using such high-dimensional data will make these systems computationally intensive and resource-heavy, rendering them unsuitable for edge deployment Li et al. (2024). Another challenge here is the vast imbalance in the captured data, which biases the modeled IDS towards the majority class Ma et al. (2025). Therefore, although recent advances have improved feature enrichment for intrusion detection, a clear gap still exists in developing an IDS framework that can jointly exploit the representational benefits of KG enrichment, maintain feature compactness, address class imbalance, and remain lightweight enough for edge-oriented deployment in IIoT.

To address these challenges, this work proposes the novel KG-Based Lightweight IDS (KGL-IDS). The KGL-IDS first uses KG-driven feature extraction to capture the underlying relationships in traffic that flow-based methods miss and join together. Then the Systematic Minority Oversampling Technique (SMOTE) was employed to balance the data. Subsequently, a hybrid feature selection filter integrating Grey Wolf Optimization (GWO) and Binary Particle Swarm Optimization (BPSO) was used, with a fitness function combining Fisher score and Correlation, to select the optimal feature set and thereby reduce dimensionality. Finally, the selected feature subset was fed into a lighter Artificial Neural Network (ANN)-based classifier for effective IDS modeling. Through this design, the proposed framework aims to balance detection capability, feature compactness, and deployment suitability for resource-constrained edge environments.

0.1 Contributions

The main contributions of this proposed KGL-IDS are:

- A lightweight IDS framework for IIoT edge is proposed by integrating KG-based feature enrichment, SMOTE-balancing, feature selection, and ANN classification.
- A practical KG-driven feature engineering mechanism is developed to transform flat flow records into contextual, relational, and interpretable tabular features using source-destination, temporal-window, protocol-like, and port-based behavioral interactions, and then combined with the flow records.
- The enriched feature space dimensionality is then reduced by using the hybrid GWO-BPSO with Fisher-Correlation fitness, hence reducing the computational overhead and enhancing the detection capabilities.
- The effectiveness of the KGL-IDS was validated through the ablation and comparative analysis using two benchmark datasets, WUSTL-IIoT and Edge-IIoT datasets.

The remainder of the article was structured as follows. Section 0.1 reviews related work on intrusion detection in IoT/IIoT edge, KG-based feature extraction, data balancing, and swarm-optimization-based feature selection. Section 0.2.1 presents the proposed KGL-IDS, its architecture, and methodology. Section 0.5.1 describes the experimental setup, dataset details, evaluation metrics, and implementation settings. Section 0.9 presents the results of the ablation study and comparative analysis on the two datasets with a discussion. Finally, Section 0.12 concludes the article with possible future directions.

RELATED WORK

There has been a significant shift towards Edge-centric IDS implementations in IoT/IIoT networks in recent years Sha et al. (2020). However, to be deployed at the edge layer, the IDS must be both lighter and more efficient. This section presents existing systems that have attempted to achieve this through different means.

0.2 Knowledge Graph-based IDS

The traditional signature-based IDS can detect known attacks, whereas the anomaly-based IDS is better suited to identifying deviations in traffic behavior, enabling it to detect new and unknown attacks as well. A flow-based IDS was proposed using a deep neural network (DNN) for an IoT environment and, when evaluated on the IoT-23 and MQTT-IoT-IDS2020 datasets, achieved F1 scores of 0.996 and 0.999, respectively Prazeres et al. (2023). Another work by Athamena et al. proposed a pruned, shallow Long Short-Term Memory (LSTM)-based IDS to enable low-power operation for edge deployment Athamena et al. (2025). IDS-IoT also employed an enhanced LSTM to detect IoT intrusions, and when tested on four datasets, it achieved accuracy exceeding 95% Meena and Indian (2025). The hybrid DLMIDPSM combined ANN, Convolutional Neural Networks (CNN), and Recurrent Neural Networks (RNN) to detect and prevent attacks in IoT and, when tested on the CICIoT2023 dataset, achieved an F1 score of 0.80 Erskine (2025).

But these approaches suffer from one major issue that is relying on predefined traffic features and often fail to explicitly capture latent semantic and relational dependencies among network entities, traffic attributes, and attack behaviors, which limits the IDS performance to an extent Lv et al. (2025). This is where KGs can help: they can extract intrinsic features hidden in the traffic that, when properly handled, can enhance IDS detection capabilities Bilot et al. (2023); Bratsas et al. (2024); Liu et al. (2022). KGs can be used to directly convert raw features into relation-based features, or to incorporate relationships into the modeling alongside existing features Bryniarska and Pokuta (2026).

The intrinsic nature of the Internet of Vehicles (IoV) makes KG-based feature representation, or extraction, highly suitable for this environment Wang et al. (2023). G-IDCS is a graph-based IDS for the in-vehicle controller area network (CAN) that demonstrated accuracies above 99% on both Car-Hacking and OTIDS datasets Park et al. (2023). A Graph Graph Convolutional Network (GCN)-ensemble fusion IDS for IoT was proposed by Mittal et al. Mittal and Batra (2024), which used a GCN to extract graph embeddings from pre-processed traffic flows and fed them to an ensemble CNN model, and, when tested on four different IDS datasets, demonstrated better performance. The HDL-IDS combined original statistical flow features with KG features and fed them into a hybrid CNN-LSTM model to detect attacks in CANs, achieving 99.44% accuracy, which is better than prior work Alqahtani and Kumar (2025). IDS-IKGAI was also proposed for intrusion detection in IoVs, converted the data into RDF triples to construct a KG, and used it to build a dataset for Random Forest (RF) training, achieving an F1 score of 0.9999 on the CICIDS2017 dataset Jin et al. (2025). The KG-ID proposed by Sun et al. Sun et al. (2025) also employed a KG-based detection methodology that demonstrated superior performance.

KG-based feature extraction and enhancement also show promising performance in general IoT networks. A lightweight IDS was proposed that uses a feature co-occurrence-based KG fused with statistical flow features to train a CNN-BiLSTM model for intrusion detection in IoT Yang et al. (2022). When tested on the NSL-KDD dataset, it achieved an F1 score of 0.9071. K-GetNID constructed heterogeneous temporal graphs from traffic flows and fed them to a heterogeneous temporal graph neural network for adjustable early detection of attacks in IoT networks, achieving 99.34% and 86.12% accuracy on CICIDS2017 and UNSW-NB15, respectively Wang et al. (2024). The TKSGF took a different approach: it used each individual traffic flow as a node, created edges based on the topmost similar neighbors, and used GraphSAGE to learn these embeddings, followed by a softmax classifier for attack detection and classification Ngo et al. (2025). When tested on the NF-ToN IoT and NF-BoT IoT datasets, the model demonstrated F1 scores of 99.99% and 98.50%, respectively.

Although existing knowledge-graph-based IDS have shown promising improvements in intrusion detection for IoV and IoT, several limitations remain. Some methods, such as TKSGF, treat each traffic record as a separate node, causing the graph to grow rapidly as traffic volume increases and reducing scalability in large-scale or streaming environments. Other approaches improve semantic representation by combining KG-derived features with statistical or raw traffic features, but this often increases dimensionality and redundancy rather than producing a compact feature space for lightweight deployment. In addition, many IoV-oriented methods rely heavily on domain-specific prior knowledge, such as CAN ontologies, DBC files, and signal-level constraints, which limit their generalizability to broader IoT settings. Moreover, several recent frameworks employ complex, multi-stage pipelines that improve accuracy but still do not sufficiently address efficiency, feature compactness, or practical edge deployment.

Therefore, although knowledge-graph-based feature enhancement improves intrusion detection by capturing semantic and relational dependencies, an important gap still remains in designing a framework that preserves these advantages while avoiding excessive graph growth, feature redundancy, and domain-specific dependence. This is particularly important for edge-oriented IIoT environments, where the IDS must remain lightweight, computationally efficient, and practically deployable.

0.2.1 Feature Optimization-based IDS

For developing an efficient IDS for IIoT networks without any domain-specific knowledge, but only with general network flow-based features, combining them with KG extracted features shows promising results, but suffers from increased dimensionality. The early IDS implementations were ML-based and hence can build not lighter but moderate models using these high-dimensional datasets. But in later years, there has been a significant shift toward DL-based IDS for IoT/IIoT Mallidi and Ramisetty (2025b). Due to the complexity of DL approaches, they were not directly suitable for deployment in edge environments without optimizing them Kantharaju et al. (2024). Hence, to achieve DL performance while maintaining low complexity and efficiency for effective anomaly detection in IIoT networks edge, recent works have explored various directions, among which feature selection-based approaches have shown promising results Bikila and Čapek (2025).

Feature selection techniques can be broadly classified into filters and wrappers. Filters use statistical approaches to find the optimal feature subset, where wrappers use the classification performance to do the same. Principal component analysis (PCA) was one widely used approach for feature selection in various ML applications, including IDS implementations. When used with various ML algorithms and evaluated on the IoTID20 dataset with 79 features, a 12-feature subset achieved the same performance as the full feature set Houkan et al. (2024). The Minimum Redundancy Maximum Relevance (MRMR) has also been implemented alongside PCA, and the 12 features selected by MRMR achieved better performance than those from PCA. Another work used both information gain and gain ratio to identify the best features and fed them to multiple ML algorithms, which, when evaluated on the IoTID20 and NSL-KDD datasets with just 28 and 25 features, achieved 99.98% accuracy Albulayhi et al. (2022).

Feature selection using both ANOVA and mutual information, together with an XGBoost, gave an AUC of 0.91 on NF-ToN-IoT-v2 data, where Kendall and Chi-square with XGBoost gave an AUC of 0.95 on NF-BoT-IoT-v2 data Hamdouchi and Idri (2025).

Combining information gain, gain ratio, correlation-based feature selection, and symmetric uncertainty together has demonstrated 96.40% accuracy, outperforming individual feature selection approaches with as few as 16 features on RT-IoT2022 Almohaimeed and Albalwy (2024). The PCA, univariate statistical test, and a Genetic Algorithm (GA) were combined to identify an optimal 10-feature set from BoT-IoT data, achieving 99.99% accuracy Mohy-eddine et al. (2023). A hybrid wrapper combining the Cellular Automata Engine and Tabu Search has identified 13 features in TON-IoT data, achieving 99.50% accuracy with RFNazir et al. (2023). Ayad et al. combined a correlation-based filter with a recursive feature elimination-based wrapper to obtain 15 features, which were used for a decision tree-based IDS construction, which has demonstrated 99.99% accuracy with both BoT-IoT and TON-IoT, and 98.74% accuracy with CICDDoS2019 Ayad et al. (2024).

Rather than a single-agent search, swarm techniques were capable of avoiding locally optimal feature sets and finding the globally optimal one. A wrapper-based feature selection using grey wolf optimization (GWO) for botnet attack detection demonstrated 99.73% accuracy with 90.10% feature reduction Penmatsa et al. (2021). Bowerbird courtship-inspired feature selection techniques were used to build a multi-level IDS for IIoT, achieving 99.70% accuracy with a simple logistic regression model Mallidi and Ramisetty (2025a). A particle swarm optimization (PSO)- driven feature selection with an ensemble classifier achieved 99.98% accuracy on CICIDS2017 and 95.20% on UNSW-NB15 Louk and Tama (2022). A modified PSO coupled with a hybrid autoencoder achieved 98.80% accuracy in attack detection on UNSW-NB15 and 99.90% on BoT-IoT Saheed et al. (2023). A hybrid wrapper combining the gorilla troops optimizer and the bird swarm algorithm demonstrated superior performance to other benchmark approaches on three IoT-IDS datasets Kareem et al. (2022).

Although feature optimization techniques have shown strong potential in reducing dimensionality and improving IDS efficiency, most existing approaches focus primarily on conventional traffic features and are not specifically designed for knowledge-graph-enhanced intrusion detection. As a result, the combined challenge of preserving the semantic benefit of KG-derived features while controlling redundancy, computational complexity, and edge deployability remains insufficiently addressed. This highlights the need for an intrusion detection framework that jointly integrates relational feature enhancement and effective feature optimization for lightweight IoT edge environments.

PROPOSED FRAMEWORK

The proposed KGL-IDS framework has three main novel phases that address gaps in the existing system and provide a lightweight IDS for IIoT by integrating KG-based feature extraction and enhancement, as shown in Figure 1. The process starts with capturing traffic, both benign and attack, from a real or lab IIoT network, extracting flow-based features, and constructing the dataset. Then, the first phase of KG-based feature enhancement that extracts knowledge graphs using traffic flow records, and the extracted KG features will be fused with the flow features. The resulting features will be pre-processed and split into training and testing datasets. Then comes the second phase, in which the SMOTE balancing technique will be used to augment the minority attack samples. Then, a hybrid feature-selection filter will be applied to this balanced train set to identify the optimal feature set, which will be used to train and evaluate an ANN IDS model.

0.3 Knowledge Graph-Based Feature Enhancement

To enhance the conventional flow-based features in the original data, a KG-based feature enhancement method was proposed to extract the underlying intrinsic relationships. In this stage, the network flows were interpreted as interactions between network entities, such as source and destination IPs, the destination port, the protocol used by the source, and, importantly, the time window during which the hosts are active, as shown in Figure 2. The source and destination hosts are the primary entities, whereas the destination ports and the protocol provide the context for their relationship. Also, each flow is mapped into a fixed-size temporal window, which was computed using Eq. (1), where t_i is the time stamp of the i^{th} flow and T_w denotes the window size.

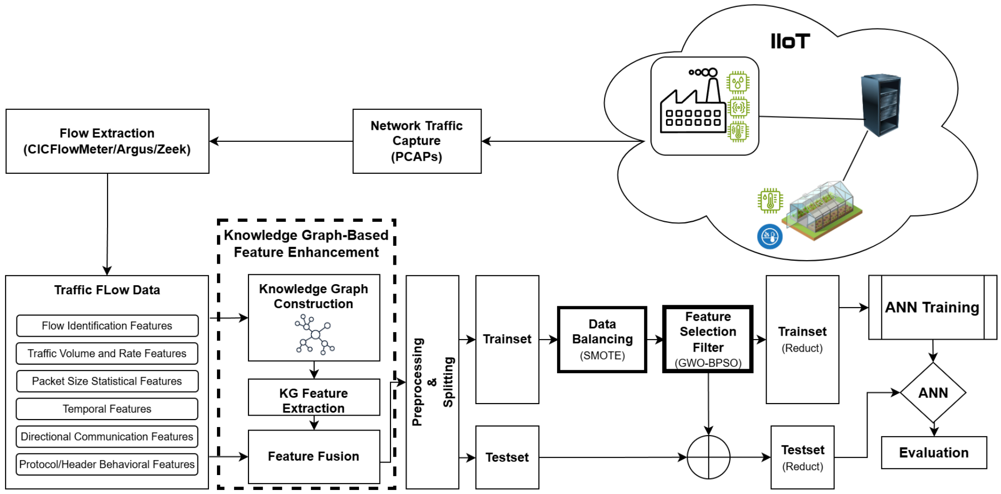

Figure 1. Architecture of Proposed KGL-IDS framework for IIoT

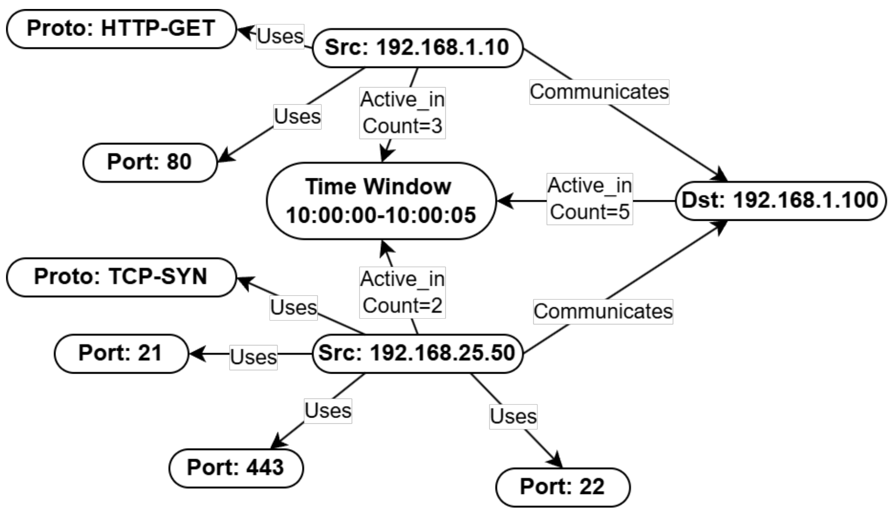

Figure 2. Knowledge graph representation of network interactions within a fixed time window

$$w_i = \left\lfloor \frac{t_i}{T_w} \right\rfloor \tag{1}$$

Instead of explicitly constructing a separate graph for each time window or avoiding heavy graph neural networks, the KG is used as an intermediate abstraction, and features are extracted by aggregating interactions among entities, as shown in Algorithm 1. The edges connecting the entities in the graph are associated with interaction frequencies, which serve as the basis for feature extraction.

[1]

Flow dataset D Fused dataset D_f

Convert timestamps into fixed-size time windows using Eq. (1) Identify entities: source host, destination host, time window, port, and protocol Construct relations among entities (communication, service usage, temporal activity) Compute structural features from source, destination, and source–destination interactions Compute temporal and diversity features using grouped aggregation Compute contextual features based on port and protocol interactions Compute ratio-based features to normalize interaction patterns Append KG-derived features to original flow features Remove identity attributes used only for KG construction **return** fused dataset D_f

Table 1. Knowledge Graph-Based Core Feature Groups and Their Behavioral Interpretation

Feature Group	Features	Behaviour Captured
Structural	src_flow_count dst_flow_count src_dst_count	Communication intensity
Temporal	time_window src_window_count dst_window_count	Time-based activity
Diversity	src_unique_dsts dst_unique_srcs src_unique_dports	Spread of communication
Contextual	src_dport_count dst_dport_count src_proto_count	Service and protocol usage
Ratio-based	src_dst_diversity_ratio src_dst_focus_ratio src_window_activity_ratio	Normalized behavior patterns

The edges connecting source and destination nodes to temporal window nodes represent the frequency of interactions within a given time interval. The interaction count between a source s and destination d is defined using $C(s,d)$ as shown in Eq. (2), and the diversity of destinations contacted by a source is given by U_s as shown in Eq. (3), where $N(s)$ represents the set of unique preliminary neighbors of s.

$$C(s,d) = \sum I(src = s \wedge dst = d) \tag{2}$$

$$U_s = |N(s)| \tag{3}$$

These edge weights directly correspond to temporal features such as src_window_count and dst_window_count. Similarly, edges between source nodes and destination ports or protocol attributes capture service usage patterns, which are aggregated to derive contextual interaction features.

A core set of KG-based features is extracted to capture structural, temporal, diversity, contextual, and ratio-based interaction patterns, as summarized in Table 1. These features model communication intensity, temporal activity, spread of interactions, and normalized behavioral characteristics.

To normalize these features and capture behavioral characteristics, ratio-based measures are computed as shown in Eq. (4) and Eq. (5), where F_s represents the total number of flows generated by the source s.

$$R_{sd}^{div} = \frac{U_s}{F_s} \tag{4}$$

$$R_{sd}^{focus} = \frac{C(s,d)}{F_s} \tag{5}$$

It is important to note that while a core set of features is consistently derived across datasets, the feature set can be further extended depending on the availability of additional network attributes. For instance, when transport-level information such as protocol types, source/destination ports, flow duration, packet counts, or byte volumes is available, additional contextual and traffic-aware features can be incorporated. Consequently, the total number of KG-derived features may vary across datasets, e.g., 23 features for Edge-IIoT and 40 features for WUSTL-IIoT, enabling the framework to adapt to different data characteristics.

Finally, the KG-derived features are fused with the original flow-based features to preserve statistical information. Identity-related attributes, such as IP addresses and timestamps, are removed after feature construction to prevent data leakage.

0.4 Data Preprocessing

In this phase, the data will undergo the typical preprocessing steps, such as normalization, handling missing values, and splitting into training and test sets. One major issue with IDS datasets is class imbalance: benign traffic and some attack traffic, such as DoS and DDoS, generate more records than attacks like XSS, SQL Injection, or brute-force attacks, which can result in a classification model biased towards the majority class.

To address this issue, SMOTE is employed. SMOTE generates synthetic samples for minority classes by interpolating between existing instances in the feature space, thereby improving class balance without simply duplicating data. Given a minority sample x_i, a synthetic sample is generated as shown in Eq. (6), where x_{nn} is one of the k nearest neighbors of x_i, and $\lambda \in [0,1]$ is a random number.

$$x_{\text{new}} = x_i + \lambda\,(x_{nn} - x_i) \tag{6}$$

In the proposed framework, SMOTE is applied only to the training dataset after the train-test split, ensuring that no synthetic information leaks into the testing phase. This preserves the integrity of model evaluation while enabling the classifier to learn more balanced decision boundaries. By integrating SMOTE into the preprocessing pipeline, the model becomes more robust in detecting minority attack classes, thereby improving overall detection performance and generalization.

0.5 Feature Selection

Feature selection plays a crucial role in developing a lightweight IDS for IIoT edge by reducing data dimensionality, thereby enhancing the IDS's efficiency and generalization capabilities. As we fuse the KG features with flow features, the data dimensionality increases further; if left unaddressed, this may negatively affect model performance.

Existing studies have highlighted that traditional feature reduction techniques, such as feature ranking, PCA, and other statistical approaches, often fail to identify optimal feature sets. Therefore, an intelligent feature selection mechanism is needed for this fused data dimensionality reduction, one that considers both relevance and redundancy while preserving the semantic richness introduced by the KG. To address these limitations, a hybrid feature selection filter was proposed by combining GWO and binary PSO, employing a hybrid fitness function that combines the feature relevance metric, Fisher score, with the redundancy metric, correlation.

Let the final feature set of fused datasets be $\mathscr{F} = \{f_1, f_2, f_3, \ldots, f_n\}$, where each agent feature subset is encoded as a binary vector $\mathbf{Z}_k = \{z_{k1}, z_{k2}, z_{k3}, \ldots, z_{kn}\}$, $z_{kj} \in \{0,1\}$, where $z_{kj} = 1$ indicates the inclusion of the feature f_j in the feature subset.

To guide the optimization process, feature relevance is computed using $Rel(\cdot)$ as shown in Eq. (7), which measures the discriminative power of a feature set $\mathbf{Z}_k$ with respect to different class labels using Fisher score. For a given feature f_j, the Fisher Score, as shown in Eq. (8), measures the ratio of inter-class variance to intra-class variance, thereby indicating how well the feature separates different classes.

$$Rel(\mathbf{Z}_k) = \sum_{j=1}^{n} z_{kj} \cdot FS(f_j) \tag{7}$$

$$FS(f_j) = \frac{\sum_{c=1}^{C} N_c \left(\mu_j^{(c)} - \mu_j\right)^2}{\sum_{c=1}^{C} N_c \left(\sigma_j^{(c)}\right)^2} \tag{8}$$

Specifically, for C classes, N_c denotes the number of samples in class c, $\mu_j^{(c)}$ represents the mean of feature f_j for class c, and μ_j is the overall mean of feature f_j across all samples. Similarly, $\sigma_j^{(c)}$ denotes the variance of feature f_j within class c. A higher Fisher score indicates that the feature exhibits large differences between class means (high inter-class separability) and low variance within each class (low intra-class variability), making it more relevant for classification. The overall relevance of a feature subset is computed by aggregating the Fisher scores of the selected features, ensuring that only features with strong discriminative capability contribute to the optimization process.

Also, to reduce the redundancy among the selected features, correlation-based redundancy is considered as shown in Eq. (9):

$$Red(\mathbf{Z}_k) = \sum_{i=1}^{n} \sum_{j=i+1}^{n} z_{ki} z_{kj} \cdot \left| corr(f_i, f_j) \right| \tag{9}$$

The overall fitness function of an agent's feature set is measured using $Fitness(\cdot)$ as shown in Eq. (10), where λ is a weighting coefficient and ε is a small constant to avoid division by zero.

$$Fitness(\mathbf{Z}_k) = \lambda \left(\frac{\sum_{j=1}^{n} z_{kj} MI(f_j, Y)}{m_k + \varepsilon} \right) - (1-\lambda) Red(\mathbf{Z}_k) \tag{10}$$

0.5.1 Hybrid GWO-BPSO Optimization

To identify the optimal feature subset, a hybrid GWO-BPSO approach is employed. GWO provides global exploration capability, while BPSO enhances local exploitation in the binary search space. The complete hybrid GWO-BPSO feature selection procedure is summarized in Algorithm 0.5.1.

The PSO updates the solution by individually updating each dimensional value of the agent's binary feature vector to z_{kj}^{t+1} as shown in Eq. (11), where $S\left(v_{kj}^{t+1}\right)$ is the probability measured using a sigmoid on the velocity of the particle in that dimension v_{kj}^{t+1}, which is updated using Eq. (12) incorporating both the best particle *Pbest* and guiding position $\mathbf{G}_k$ of GWO.

$$z_{kj}^{t+1} = \begin{cases} 1, & \text{if } S\left(v_{kj}^{t+1}\right) \geq 0.5 \\ 0, & \text{Otherwise} \end{cases} \tag{11}$$

$$v_{kj}^{t+1} = w\, v_{kj}^{t} + c_1 r_1 \left(Pbest_{kj} - z_{kj}^{t} \right) + c_2 r_2 \left(G_{kj} - z_{kj}^{t} \right) \tag{12}$$

The GWO gives guiding position $\mathbf{G}_k$ that is measured using Eq. (13), which is a relative position measured from the three best wolves, i.e., $\mathbf{X}_\alpha$, $\mathbf{X}_\beta$, and $\mathbf{X}_\delta$, as guided by the coefficient and distance vectors as shown in Eq. (14), Eq. (15), and Eq. (16).

$$\mathbf{G}_k = \frac{\mathbf{X}_1 + \mathbf{X}_2 + \mathbf{X}_3}{3} \tag{13}$$

$$\mathbf{X}_1 = \mathbf{X}_\alpha - \mathbf{A}_1 \cdot \mathbf{D}_\alpha \tag{14}$$

$$\mathbf{X}_2 = \mathbf{X}_\beta - \mathbf{A}_2 \cdot \mathbf{D}_\beta \tag{15}$$

$$\mathbf{X}_3 = \mathbf{X}_\delta - \mathbf{A}_3 \cdot \mathbf{D}_\delta \tag{16}$$

where, the distance between a candidate solution and a leader solution is defined as $\mathbf{D} = |\mathbf{C} \cdot \mathbf{X}_l - \mathbf{Z}_k|$, with $\mathbf{X}_l \in \{\mathbf{X}_\alpha, \mathbf{X}_\beta, \mathbf{X}_\delta\}$, and $\mathbf{A}$ and $\mathbf{C}$ are coefficient vectors updated in every iteration using the diminishing a as shown in Eq. (17) and Eq. (18).

$$a = 2 - 2\left(\frac{t}{T}\right) \tag{17}$$

$$\mathbf{A} = 2a\mathbf{r}_1 - a, \quad \mathbf{C} = 2\mathbf{r}_2 \tag{18}$$

[1]

Fused dataset D_f with feature set $\mathscr{F}$, population size P, iterations T Optimal feature subset $\hat{\mathscr{F}}$

Compute $FS(f_j)$ for all features $f_j \in \mathscr{F}$ Initialize binary populations $\mathbf{Z}_k$ and velocities $\mathbf{V}_k$ Evaluate the fitness of each agent's binary feature vector $\mathbf{Z}_k$ using Eq. (10) Identify the top 3 agents' solutions and mark them as: $\mathbf{X}_\alpha$, $\mathbf{X}_\beta$, and $\mathbf{X}_\delta$

$t = 1$ to T Update a using Eq. (17) each agent k Compute GWO guiding position $\mathbf{G}_k$ using Eq. 13 each feature dimension j Update the velocity using Eq. (12) Update the binary position using Eq. (11) Evaluate the fitness of all agents using Eq. (10) Update personal best *Pbest* if improved Update $\mathbf{X}_\alpha$, $\mathbf{X}_\beta$, and $\mathbf{X}_\delta$ Update the global best feature set $\hat{\mathscr{F}}$ if improved

return best feature subset $\hat{\mathscr{F}}$

The proposed hybrid GWO-BPSO feature selection approach effectively balances exploration and exploitation, enabling the identification of a compact and highly discriminative feature subset. By incorporating both statistical relevance and KG-derived semantic features, the method enhances detection performance while reducing computational complexity. This makes the framework suitable for deployment in resource-constrained IIoT edge environments.

After feature selection, the reduced and optimized feature subset is used to train a classification model for intrusion detection. In this work, a feedforward ANN is employed because it can model complex nonlinear relationships between features and class labels while maintaining a relatively lightweight architecture suitable for IIoT environments. The selected features, comprising both conventional flow-based attributes and KG-derived features, provide a rich representation that the ANN effectively exploits for accurate classification.

EXPERIMENTAL SETUP

This section presents the datasets, preprocessing pipeline, feature engineering, feature selection configuration, model implementation, and evaluation protocol for validating the proposed KGL-IDS framework.

0.6 Datasets

The proposed model is evaluated on two publicly available Industrial IoT datasets: the Edge-IIoT and WUSTL-IIoT datasets. These datasets are selected due to their realistic representation of IIoT environments and diverse attack scenarios.

The Edge-IIoT dataset consists of large-scale network traffic collected from an edge-enabled IoT environment and contains both benign traffic and multiple attack categories, including DDoS/DoS, reconnaissance, brute-force, web attacks, and malware. The dataset includes flow-level features extracted from raw network traffic, making it suitable for evaluating edge-based intrusion detection systems.

The WUSTL-IIoT dataset captures industrial control system traffic and includes detailed flow statistics, such as packet and byte counts, timing features, and protocol attributes. It contains both normal and malicious traffic, enabling robust evaluation in industrial IoT scenarios.

0.7 Knowledge Graph Feature Construction

To capture semantic relationships among network entities, a KG-inspired feature engineering process is applied. Network attributes such as source IP, destination IP, ports, and protocols are treated as entities and relationships, and interaction patterns are modeled using temporal windows.

Specifically, time-window-based grouping (5-second intervals) is used to capture temporal behavior, and features such as interaction counts, unique connection diversity, port usage patterns, and activity ratios are extracted. These include features such as source/destination flow counts, unique destination counts, protocol diversity, and window-based activity metrics. The resulting knowledge-graph-derived features are combined with conventional flow-based features to create a semantically enriched feature space, thereby improving the representation of network behavior.

0.8 Feature Selection Setup

The parameter settings used in the hybrid GWO-BPSO feature selection are summarized in Table 2. These values are selected based on commonly adopted configurations in metaheuristic optimization and further refined through empirical evaluation to balance exploration, exploitation, and computational efficiency. A moderate population size of 30 and 50 iterations is chosen to ensure sufficient search capability while maintaining a manageable computational cost. The GWO control parameter is linearly decreased from 2

Table 2. Parameter Settings for Hybrid GWO-BPSO

Parameter	Value
Population size	30
Maximum iterations	50
Inertia weight w range	$[0.4, 0.9]$
Cognitive and Social Coefficients c_1 & c_2	1.8
Initial position range	$[-2, 2]$
Initial velocity range	$[-0.5, 0.5]$
Position clipping range	$[-6, 6]$
Fitness trade-off parameter λ	0.995

to 0, gradually transitioning the search process from exploration to exploitation. Similarly, the inertia weight w in the BPSO component is reduced from 0.9 to 0.4 to encourage global exploration in the initial stages and local refinement in later iterations.

The acceleration coefficients c_1 and c_2 are set equally to maintain a balanced influence between individual learning and global guidance. The initialization ranges for positions and velocities ensure diversity in the initial population, while clipping bounds are applied to maintain numerical stability during optimization. Finally, the fitness trade-off parameter λ is set to a high value to prioritize feature relevance while still penalizing redundancy.

0.9 Model Configuration & Evaluation

The classification model is implemented using a feedforward ANN. The network consists of three hidden layers with 256, 128, and 64 neurons, respectively. Each layer is followed by batch normalization, ReLU activation, and dropout regularization to improve generalization and prevent overfitting. The output layer uses a Softmax activation function for multiclass classification. The model is optimized using the Adam optimizer with a learning rate of 10^{-3} and trained using the categorical cross-entropy loss function. Additionally, learning rate reduction is applied when the validation loss plateaus, ensuring stable convergence during training with a minimum learning rate of 10^{-6}.

The performance of the proposed KGL-IDS framework is evaluated using standard classification metrics, including Accuracy, Precision, Recall, and F1-score, to provide a comprehensive evaluation. In addition, confusion matrix analysis and detailed classification reports are used to assess class-wise detection performance and identify misclassification patterns.

RESULTS & DISCUSSION

This section presents a comprehensive evaluation of the proposed KGL-IDS framework on the Edge-IIoT and WUSTL-IIoT datasets. The results are analyzed through ablation studies, feature selection analysis, and comparative evaluation with existing benchmark approaches to demonstrate the effectiveness of the proposed method.

0.10 Ablation Study

To assess the contribution of each component in the proposed framework, an ablation study is conducted using four configurations: (i) baseline ANN on the original network flow dataset, (ii) ANN with the KG enhanced dataset, and (iii) the full KGL-IDS incorporating KG feature enhancement, SMOTE-based data balancing, and hybrid GWO-BPSO feature selection.

Table 3 and Table 4 present the class-wise accuracy and class-wise F1 score, respectively, for the Edge-IIoT dataset. The Edge-IIoT results show that KG-based feature enrichment consistently improves minority-class recognition compared with the baseline ANN. In particular, the KG configuration improves the F1-score for the Recon and Brute Force classes while maintaining perfect or near-perfect detection across most classes. The proposed KGL-IDS configuration achieves the best overall balance, preserving strong class-wise performance while improving the macro F1-score and reducing the feature space to 27 selected features.

Table 5 and Table 6 summarize the class-wise F1 score and accuracy, respectively, for the WUSTL-IIoT dataset. For the WUSTL-IIoT dataset, the benefit of KG-based feature enrichment is even more

Table 3. Class-wise Accuracies for Edge-IIoT

Class	Baseline	KG	KGL-IDS (Proposed)
Normal	1.00	1.00	1.00
Backdoor	0.97	0.97	0.97
Brute Force	0.98	0.99	1.00
DDoS DoS	0.99	1.00	1.00
Malware	0.89	0.90	0.97
Recon	0.99	1.00	1.00
Web Attacks	1.00	1.00	1.00

Table 4. Class-wise F1 Score for Edge-IIoT

Class	Baseline	KG	KGL-IDS (Proposed)
Normal	1.00	1.00	1.00
Backdoor	0.98	0.98	0.98
Brute Force	0.99	1.00	1.00
DDoS DoS	0.99	1.00	1.00
Malware	0.94	0.94	0.97
Recon	0.96	0.99	0.99
Web Attacks	0.99	1.00	1.00

evident. Although the baseline ANN achieves very high overall accuracy, its class-wise performance on minority attack classes such as Backdoor and CommInj is noticeably weaker. After incorporating KG-derived features, both class-wise F1-score and class-wise accuracy improve substantially for these classes. The proposed KGL-IDS maintains this strong class balance while reducing the feature space to 23 selected features, demonstrating that the hybrid GWO-BPSO feature selection preserves the most informative semantic and statistical features without sacrificing detection capability.

Overall, the ablation results confirm that each component of the proposed framework contributes meaningfully to the final performance. KG-based features enhance relational representations, and hybrid GWO-BPSO feature selection produces a compact yet highly discriminative subset that supports robust, lightweight IIoT intrusion detection. Even though the improvements seem minimal, they are very significant in IDS-like security applications.

0.11 Feature Selection Effectiveness and Stability

The effectiveness of the proposed hybrid GWO-BPSO feature selection is further analyzed with respect to dimensionality reduction and model stability. Table 7 shows the number of features in the original flow-based dataset, the enriched feature space obtained after KG fusion, and the final subset selected by the proposed approach. It can be observed that KG-based enrichment increases the dimensionality of the feature space, thereby enhancing the model's representational capacity. However, this increase also introduces redundancy and computational overhead. The proposed hybrid GWO-BPSO feature selection effectively addresses this issue by selecting only the most informative features, thereby reducing the final feature space to 27 for Edge-IIoT and 23 for WUSTL-IIoT.

To evaluate the impact of feature selection on model robustness, the distributions of Accuracy and F1-score across multiple runs are shown in box plots in Figures 3 and 4. The comparison is performed between the model without feature selection (KG+SMOTE) and the proposed model with hybrid GWO-BPSO feature selection.

For the Edge-IIoT dataset (Figure 3) and the WUSTL-IIoT dataset (Figure 4), both Accuracy and F1-score exhibit relatively narrow distributions due to the dataset's inherent strong separability. Nevertheless, the model without feature selection shows slightly wider interquartile ranges and minor extreme variations. In contrast, the proposed model exhibits a tighter distribution, indicating improved consistency and reduced sensitivity to redundant features.

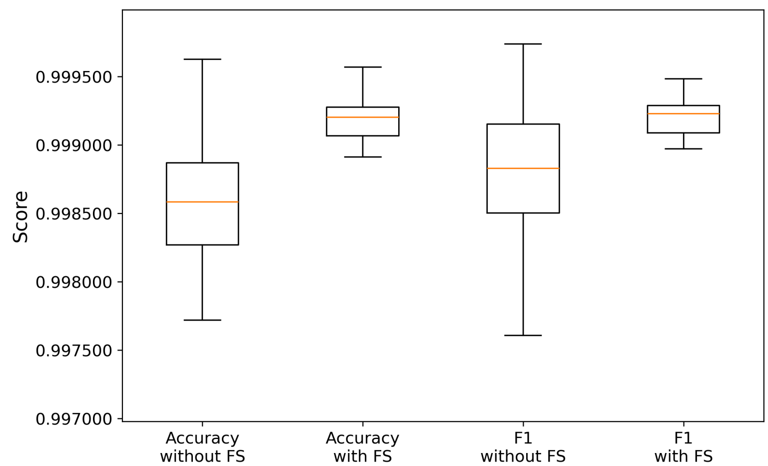

Figure 3. Box plot comparison of Accuracy and F1-score on Edge-IIoT dataset with and without feature selection

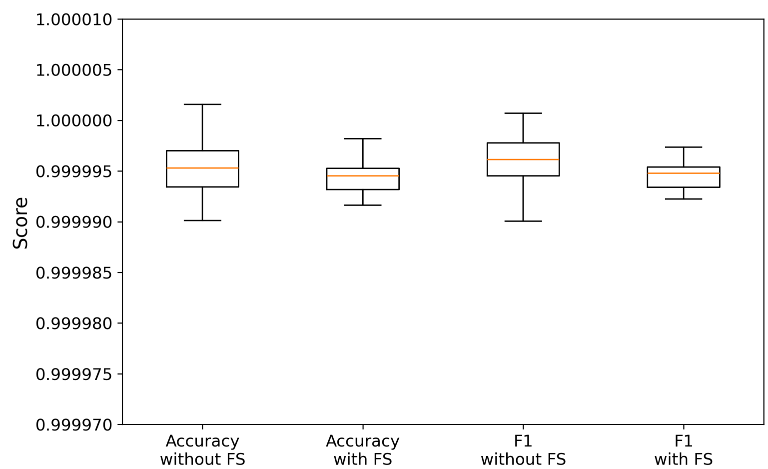

Figure 4. Box plot comparison of Accuracy and F1-score on WUSTL-IIoT dataset with and without feature selection

Table 5. Class-wise Accuracies for WUSTL-IIoT

Class	Baseline	KG	KGL-IDS (Proposed)
Normal	1.00	1.00	1.00
Backdoor	0.74	1.00	1.00
CommInj	0.90	1.00	1.00
DoS	1.00	1.00	1.00
Recon	1.00	1.00	1.00

Table 6. Class-wise F1 Score for WUSTL-IIoT

Class	Baseline	KG	KGL-IDS (Proposed)
Normal	1.00	1.00	1.00
Backdoor	0.83	0.99	1.00
CommInj	0.91	0.99	0.99
DoS	1.00	1.00	1.00
Recon	1.00	1.00	1.00

0.12 Comparison with Existing Methods

To further validate the effectiveness of the proposed approach, KGL-IDS is compared with several benchmark methods, including traditional machine learning, graph-based approaches, and deep learning-based intrusion detection models. Since the compared works use different sampling strategies, train-test splits, and experimental settings, the comparison is intended to provide a practical performance perspective rather than a perfectly controlled one. Nevertheless, it still provides useful insight into the relative effectiveness of the proposed framework.

Table 8 presents the comparative results on the Edge-IIoT dataset. The proposed KGL-IDS also achieves the best performance on Edge-IIoT, outperforming all compared methods in both overall accuracy and macro F1-score. This improvement is particularly important because macro F1-score reflects balanced recognition of minority classes such as Malware, Backdoor MITM, and Recon. Compared with graph-based methods, the proposed framework demonstrates that combining KG-based semantic feature enrichment with compact hybrid feature selection is more effective than relying solely on graph modeling.

Table 9 compares the proposed KGL-IDS with representative benchmark methods on the WUSTL-IIoT dataset. The results show that the proposed KGL-IDS achieves the strongest overall performance on WUSTL-IIoT. The proposed framework substantially improves macro F1-score, demonstrating that KG-based feature enrichment, SMOTE-based balancing, and hybrid GWO-BPSO feature selection together provide a much better balance across all attack categories.

CONCLUSION AND FUTURE WORK

This proposed KGL-IDS, a KG-based lightweight intrusion detection framework for IIoT environments that integrates KG-driven feature enhancement, SMOTE-based data balancing, hybrid GWO-BPSO feature selection, and ANN-based classification. The proposed framework addresses key challenges in IIoT intrusion detection, including the lack of relational representation in flow-based features, severe class imbalance, and the increase in dimensionality after feature enrichment. By transforming raw flow data into interaction-aware KG features and subsequently selecting a compact subset using hybrid optimization, the framework achieves a balance between representational richness and computational efficiency. Experimental results on the Edge-IIoT and WUSTL-IIoT datasets demonstrate that KGL-IDS achieves high accuracy and macro F1-score while maintaining strong class-wise performance, particularly for minority attack categories. The ablation, feature reduction, and comparative analyses confirm that each component contributes meaningfully to improving detection capability, model stability, and feature compactness, making the framework suitable for deployment in resource-constrained edge environments.

Despite these promising results, the present work has certain limitations. The evaluation is conducted on offline benchmark datasets under controlled experimental conditions, and the current framework assumes a centralized training setup. In real-world IIoT scenarios, network conditions are dynamic, and data distributions may evolve over time, which may affect model generalization. Furthermore, while the

Table 7. Feature Reduction Analysis

Dataset	Original	KG Enhanced	HGWO-BPSO
Edge-IIoT	56	79	27
WUSTL-IIoT	44	84	23

Table 8. Comparative Analysis on Edge-IIoT

Method	Accuracy	F1 (Macro)
Mittal & Batra (2024) Mittal and Batra (2024)	0.9554	0.9058
Wang et al. (2024) Wang et al. (2024)	0.9964	0.9798
Yang et al. (2022) Yang et al. (2022)	0.8854	0.6662
Park et al. (2023) Park et al. (2023)	0.8341	0.4497
Ngo et al. (2025) Ngo et al. (2025)	0.9842	0.9132
KGL-IDS (Proposed)	0.9992	0.9919

ANN classifier provides a lightweight solution, it does not explicitly capture temporal dependencies or distributed learning characteristics that may be present in large-scale IIoT deployments.

As future work, the proposed KGL-IDS framework can be extended in several directions. First, its performance can be validated in real-time or streaming IIoT environments to assess deployment feasibility under practical constraints. Second, the framework can be adapted to federated or distributed learning settings to enhance scalability and privacy across multiple edge nodes. Third, adaptive or dynamic KG construction strategies can be explored to better capture evolving attack behaviors. Finally, further optimization of lightweight models and hardware-aware implementations can be investigated to reduce computational overhead and improve energy efficiency for real-world edge deployments.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used OpenAI's ChatGPT for language enhancement and readability improvement. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Table 9. Comparative Analysis on WUSTL-IIoT

Method	Accuracy	F1 (Macro)
Mittal & Batra (2024) Mittal and Batra (2024)	0.9968	0.8899
Wang et al. (2024) Wang et al. (2024)	0.9959	0.9409
Yang et al. (2022) Yang et al. (2022)	0.9999	0.9538
Park et al. (2023) Park et al. (2023)	0.9861	0.7513
Ngo et al. (2025) Ngo et al. (2025)	0.9996	0.9134
KGL-IDS (Proposed)	0.9999	0.9962

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